数学理论推导

01 / 模型与弱形式 - T01-T02 - 主线任务

清单使用说明
Each task states its methods, approach, and deliverable. T01-T08 cover the core problem; T09-T12 are extensions; T13 is the optional pure-Neumann study. Code tasks use C / X identifiers. The added solver tasks continue in T14-T20.

[ ] T01 明确方程、边界与制造解

使用方法：Manufactured solutions; direct differentiation; outward-normal checks.

如何完成：Write −Δu=f on the unit square. Assign Dirichlet data to the left, bottom, and top edges, including all four corners. On the open right edge, study Neumann and Robin data separately, with β=2. Differentiate the exact solution to verify the data below and the outward normal n=(1,0).

完成标志：A model-data table and boundary diagram that explicitly assign every corner to Dirichlet data.

$$u_*=e^{x+y},\ f=-2e^{x+y},\ g=u_*|_{\Gamma D}$$
$$q=e^{1+y},\ r=3e^{1+y}=q+2u_*(1,y)$$

[ ] T02 推导弱形式并说明唯一性

使用方法：Green's identity; the energy argument; the Poincaré inequality.

如何完成：Choose test functions with zero trace on the Dirichlet edges and integrate by parts. Put the prescribed Neumann flux on the right-hand side and the Robin βuv term on the left. Subtract two candidate solutions, then test with their difference to establish uniqueness using the nonempty Dirichlet boundary.

完成标志：One weak formulation for each boundary variant, with an explanation of why pure-Neumann compatibility is unnecessary here.

$$a(v,v)=\int_\Omega|\nabla v|^2+\int_{\Gamma B}\beta v^2 \geq \|\nabla v\|^2_{L^2}$$

Set β=0 for Neumann data. Coercivity concerns a(v,v), not a lower bound for arbitrary a(u,v).

内部与一阶边界离散

02 / 五点方程与导数行 - T03-T04 - 主线任务

[ ] T03 推导内部格式并消去已知值

使用方法：Centered second differences; Taylor expansion; algebraic elimination.

如何完成：Set h=1/N, x_i=ih, and y_j=jh. Approximate each second derivative by a centered difference and expand the result in Taylor series. Combine the two directions to obtain the five-point equation. Whenever a neighboring value is prescribed by Dirichlet data, move its contribution to the right-hand side with the correct sign.

完成标志：The interior row, its O(h²) local defect, and the +g/h² rule for known neighbors.

$$(4U_{i,j}-U_{i-1,j}-U_{i+1,j}-U_{i,j-1}-U_{i,j+1})/h^2=f_{i,j}$$

Unknowns satisfy 1≤i≤N and 1≤j≤N−1, so m=N(N−1). Assemble interior PDE rows only for 1≤i,j≤N−1.

[ ] T04 推导一阶诺伊曼与罗宾边界行

使用方法：One-sided backward differences; strong enforcement of boundary equations.

如何完成：Approximate the outward derivative on the open right edge using a backward difference. Insert this expression into the Neumann and Robin conditions separately. Each resulting derivative equation replaces the PDE row at that boundary node. Keep the boundary value as an unknown, and do not also assemble a PDE equation there.

完成标志：Two assembly-ready boundary rows, with diagonal coefficients, inward couplings, and right-hand sides identified.

$$\text{Neumann: } (U_{N,j}-U_{N-1,j})/h=q_j$$

$$\text{Robin: } (1/h+\beta)U_{N,j}-U_{N-1,j}/h=r_j$$

对应代码任务 C02-C03

All corners remain prescribed Dirichlet values. Right-edge rows apply only for j=1,…,N−1. The column i=N−1 is still interior and must remain connected to the unknown right-edge values.

虚点消元与公式核验

03 / 虚点仅用于推导 - T05 - 主线任务

[ ] T05 推导虚点消元行并检查局部缺陷

使用方法：Centered normal differences; ghost elimination; Taylor expansion.

如何完成：Express the exterior value $U_{N+1,j}$ through the centered boundary condition, then substitute it into the five-point PDE at the right edge. The inward coupling doubles, and no ghost unknown remains. Substitute exact nodal samples into every row type to check signs, powers of h, and leading defect terms.

完成标志：Two ghost-eliminated boundary PDE rows and substitution checks for both closure types.

$$U_{N+1,j}=U_{N-1,j}+2hq_j \text{ (Neumann)}$$

$$U_{N+1,j}=U_{N-1,j}+2h(r_j-\beta U_{N,j}) \text{ (Robin)}$$

$$(4U_{N,j}-2U_{N-1,j}-U_{N,j-1}-U_{N,j+1})/h^2 = f_{N,j}+2q_j/h$$

$$(4/h^2+2\beta/h)U_{N,j}-2U_{N-1,j}/h^2 -(U_{N,j-1}+U_{N,j+1})/h^2=f_{N,j}+2r_j/h$$

精确解代入：左端减右端	本制造解的缺陷首项
内部 PDE 行	$-h^2u_{xxxx}/6+O(h^4)$
一阶边界行（两种条件）	$-hq_j/2+O(h^2)$
虚点边界 PDE 行（两种条件）	$hq_j/3+O(h^2)$

区分局部与全局误差阶： the centered derivative formula is second order, but the eliminated, unscaled boundary PDE defect is O(h). Multiplying that row by h² gives O(h³). Neither local statement alone determines the expected second-order global error.

The comparison also changes how equations are imposed: a derivative row replaces the PDE, whereas the ghost closure retains a boundary PDE after elimination. See code tasks C03-C04.

误差度量与矩阵诊断

04 / 先定义度量，再解释结果 - T06-T07 - 主线任务

[ ] T06 定义分区误差与收敛阶计算规则

使用方法：Discrete L² norms; nodal L∞ norms; adjacent-grid rates; log-log fitting.

如何完成：Let $e=U-u_*$. Fix the global node set S, interior set I, and open-right-edge set B. Use area weights for global and interior L² errors, and line weights on the boundary. Run at least four meshes per case, identify an asymptotic range, and only then fit convergence rates.

完成标志：A norm-definition table, adjacent-grid rates, and an explicit rule identifying the meshes used for fitting.

$$E_{2,h}=\sqrt{(h^2\Sigma_S e^2)},\ E_{2,I}=\sqrt{(h^2\Sigma_I e^2)},\ E_{2,B}=\sqrt{(h\Sigma_B e^2)}$$

$$E_\infty=\max|e| \text{ on each set};\ p=\log(E_h/E_{h/2})/\log 2$$

S: $1\le i\le N$, $1\le j\le N-1$. I: $1\le i,j\le N-1$. B: $i=N$, $1\le j\le N-1$. Expected: first-order boundary $E_{\infty,B}$ approaches order one; ghost global errors approach order two in both norms. Verify these expectations experimentally.

[ ] T07 解释非对称性并统一缩放残差

使用方法：Sparse structure; the Frobenius norm; row scaling.

如何完成：Compare boundary-to-interior and reverse coupling coefficients to explain why the original A is generally nonsymmetric. Report its relative nonsymmetry and define a residual normalization from absolute row sums. Use exactly the same evaluation function across meshes and solvers, including methods that internally solve an equivalent scaled system rather than the original equations.

完成标志：A sparse-pattern plot, $\delta_A=\|A-A^T\|_F/\|A\|_F$, and one common residual definition.

$$(S_r)_{kk}=1/\Sigma_\ell|A_{k\ell}|$$

$$\rho_s=\|S_r(b-AU)\|_\infty/(\|S_rA\|_\infty\|U\|_\infty+\|S_rb\|_\infty)$$

S_r normalizes the residual; it is distinct from the SPD row weight W in T14. If numerator and denominator vanish, report zero; if only the denominator vanishes, report infinity. Investigate zero rows. Any condition number must specify its norm and scaling; do not assume a universal h^{-2} law.

同精度求解器比较

05 / 正确更新、停止准则与解释 - T08 - 主线任务

[ ] T08 比较直接法基准与三种定常迭代

使用方法：Sparse direct solution; matrix splitting; iteration-matrix spectral radii.

如何完成：Use a sparse direct solution as the reference. Write A=D−E−F, where E and F are the negatives of the strictly lower and upper parts of A, and derive the updates below. Do not apply CG directly to the original nonsymmetric matrix; T14 constructs its equivalent SPD system for the added solvers.

完成标志：Correct iteration formulas, a common stopping rule, an accuracy budget, and a fair timing protocol.

$$\text{Jacobi: } U^{k+1}=D^{-1}[b+(E+F)U^{k}]$$

$$\text{GS: } (D-E)U^{k+1}=b+FU^{k}$$

$$\text{SOR: } (D/\omega-E)U^{k+1}=[(1/\omega-1)D+F]U^{k}+b$$

怎样匹配求解精度

Recompute the true scaled residual each sweep; start with tol=10^{-10}. For every case and mesh, also require $\|U_{it}-U_{dir}\|_{\infty}\le 0.01E_{\infty,dir}$. Tighten the tolerance if needed. A fixed residual threshold alone does not guarantee this 1% error budget.

说明主题	具体要求
收敛性	Check or justify the iteration-matrix spectral radius. For nonsymmetric systems, 0<ω<2 alone does not guarantee SOR convergence.
停止与失败处理	Stop on the residual; set a maximum iteration count and retain nonconvergence states.
网格与计算量	Model reference: iteration-count slopes versus N are about +2, +2, +1. Measure the mixed-boundary results.
时间与交叉点	Use a fixed timing protocol; test for a crossover without assuming a ranking or its absence.

Nonzero sources and fluxes invalidate a universal claim that the solution lies between boundary extrema. If using a comparison principle, construct bounds incorporating the data. See code tasks C05-C08; the four added methods are developed in T14-T20.

FEM 对照与 L 形域

06 / 混合边界研究的独立拓展 - T09-T10

[ ] T09 建立矩形 FDM / FEM 对照

使用方法：P1 shape functions; the Galerkin method; element assembly.

如何完成：Define a separate smooth Poisson problem on the square with Dirichlet data on its entire boundary. Use shared grid nodes for five-point FDM and P1 triangles. Derive element stiffness matrices, then match mesh scale, error norms, and solver accuracy before comparing stiffness, load construction, assembly time, and solution cost.

完成标志：A fair-comparison specification; on this regular right-triangle mesh, interior stiffness satisfies $A_{FEM}=h^2A_{FDM}$.

Stiffness couplings along the square diagonals vanish: the effective interior stencil still has five points, not seven or nine nonzero entries. Load vectors may differ, so the solutions need not coincide.

[ ] T10 解释角点奇异性与收敛阶退化

使用方法：The polar Laplacian; Sobolev regularity; Céa's lemma; duality.

如何完成：Use $\Omega=(-1,1)^2\setminus[0,1)^2$, $f=0$, and $g=u_*$ on the whole boundary. Implement the explicit atan2 branch correction and verify harmonicity in polar coordinates. Explain how limited regularity restricts P1 FEM error estimates. Treat FDM separately by defining its discrete energy norm and labeling rates as predictions or measured results.

完成标志：Branch and boundary checks, norm-specific FEM predictions, and a separate interpretation of FDM rates.

$$u_*=r^{2/3}\sin(2\varphi/3),\ \varphi=\theta-\pi/2$$

$$\theta=\operatorname{atan2}(y,x);\ \text{if } \theta<\pi/2,\ \text{set } \theta\leftarrow\theta+2\pi$$

$u_*\in H^{1+s}$ for every $s<2/3$, but generally $u_*\notin H^2$. Quasi-uniform P1 FEM predicts energy and L^2 rates near 2/3 and 4/3, with arbitrarily small exponent losses in a strict statement. This proof cannot be applied directly to FDM.

Assign $u_*(0)=0$ without evaluating its gradient there. The squared gradient behaves as $r^{-2/3}$ and is integrable in two dimensions. Retain corner elements in energy-error integration. See code tasks X01-X02.

加密策略与兼容性

07 / 三项独立拓展任务 - T11-T13

[ ] T11 设计分级网格

使用方法：Interpolation-error scaling; shape-regular local refinement.

如何完成：Start from the corner size law $h_K \approx H\, r_K^{1-1/\mu}$, with core size about H^{μ}. Choose $\mu > 3/2$, initially 2, and explain boundary preservation and element quality. Prefer local refinement of an L-shaped triangulation. If retaining a radial construction, supply the outer-boundary mapping and control angular resolution as well as radial spacing.

完成标志：A size law and mesh rule; measured energy-error slopes versus degrees of freedom, without assuming optimal recovery.

The critical value $\mu = 3/2$ generally has a logarithmic loss. A radial construction needs $R(\theta) = 1/\max(|\cos\theta|, |\sin\theta|)$. An anisotropic mesh cannot inherit a shape-regular proof without further analysis.

[ ] T12 定义自适应指标与标记规则

使用方法：Residual estimation; normal-flux jumps; conforming refinement.

如何完成：Define cell residuals and normal-flux jumps across interior edges, then run the solve-estimate-mark-refine loop. Use Dörfler marking with a fixed parameter θ. Choose reliable newest-vertex bisection or red-green refinement, specify conformity closure, and compare with uniform refinement using identical error norms, quadrature accuracy, and algebraic stopping criteria.

完成标志：An indicator, marking parameter, closure rule, and comparison protocol using whole-domain continuous L^2 and energy integrals.

$$\eta_K^2 = h_K^2 \|f + \Delta u_h\|_{0,K}^2 + (1/2) \sum_{E \subset \partial K,\ \text{interior}} h_E \|[[\nabla u_h \cdot n_E]]\|_{0,E}^2$$

For P1 and $f = 0$, the cell term vanishes, but normal jumps may remain. Do not substitute tangential jumps. Integrate over corner elements using suitable quadrature; record how nonhomogeneous boundary data are approximated.

[ ] T13 处理纯诺伊曼的兼容条件与规范

使用方法：The divergence theorem; left and right nullspaces; gauge constraints.

如何完成：Prescribe flux on every boundary edge and first verify $\int_\Omega f + \int_{\partial\Omega} q = 0$. Discretely, check $A1 \approx 0$ and a left null vector $w^T A \approx 0$. Enforce $w^T b = 0$ before fixing the mean or one node. Shift the exact solution to the same gauge before measuring error, and record any compatibility correction to the right-hand side.

完成标志：Continuous and discrete compatibility checks, the constant null vector, and a documented gauge; setting an entire edge to zero is not a gauge.

Keep extensions separate. All outputs are planned; T14 explains symmetric gauges for CG.

为什么这样设计比较实验

实验设计说明 / 进入 T14-T20 前先读本页

实验目标： Make each solver reach the same solution accuracy, then compare the time, memory and robustness needed to get there. Vary the matrix, mesh, size and number of right-hand sides to identify suitable use cases. This is an experimental plan, not a measured ranking.

系统	可用方法	分组依据
原始 AU=b	Direct, Jacobi, GS, SOR and GMRES	Preserve the prescribed nonsymmetric assembly.
等价 BU=c	All seven independent methods, plus the direct reference and two PCG combinations	Use one shared SPD matrix for a fair comparison that includes CG.

Here B=WA and c=Wb. Multiplying both sides of each equation by a specified positive weight preserves its solution. T14 derives the weights and proves that this particular B is symmetric positive definite (SPD). Standard CG is not applied directly to the original A.

> 十类对比对象与多重网格的两种角色
> The seven independent methods are Jacobi, GS, SOR, CG, GMRES, standalone AMG and standalone GMG. Add a sparse direct reference, CG+AMG and CG+GMG: ten labels in total. Standalone MG repeatedly applies V-cycles; PCG uses one fixed V-cycle as each preconditioner application.

Keep the original A,b available even when solving B. Accuracy is assessed against that original equation. A FEM stiffness system K uses K,d as its original equation. The comparison changes the solver without silently changing the discrete problem.

阅读顺序： T14 establishes the common platform; T15-T19 explain the methods and compatibility conditions; T20 connects the four scenarios S1-S4 to an evidence-based choice. Implementation tasks C09-C14 use the same notation.

CG 的等价对称正定系统

T14 / 求解器比较必做任务

[ ] T14 推导行权重并证明正定性

使用方法：Positive row scaling; weighted graph energy; Dirichlet constraints.

如何完成：Retain the original row scales from T03-T05. Left-multiply A and b by the positive diagonal matrix W below. Check paired coupling coefficients, then prove symmetry and positive definiteness through the weighted graph energy. Arbitrary positive row scaling does not generally produce an SPD matrix.

完成标志：A weight table, an equivalence argument, a symmetry check and an SPD proof.

行类型 / 边界闭合	对角权重 W	缩放后的向内耦合
内部 PDE 行	1	$-1/h^2$
一阶右边界行	1/h	$-1/h^2$
虚点右边界行	1/2	$-1/h^2$

$$AU=b \Leftrightarrow BU=c,\ B=WA,\ c=Wb$$

$$z^{T}Bz=\Sigma_{k<\ell}\gamma_{k\ell}(z_k-z_\ell)^2+\Sigma_k d_k z^2{}_k$$

Let $\gamma_{k\ell}=-B_{k\ell}\geq 0$ and $d_k=B_{kk}-\Sigma_{\ell\neq k}\gamma_{k\ell}\geq 0$. The graph is connected, and rows adjacent to eliminated Dirichlet data have positive surplus. For $\beta\geq 0$, the energy is strictly positive for every nonzero z.

证明的适用范围

These weights assume the stated uniform grid, Dirichlet elimination and original row scales. Re-derive them if rows were already multiplied by h or h^2. CG solves B, while final residuals are checked with A,b.

A nonempty Dirichlet boundary is essential here. A pure-Neumann system remains singular before a gauge; a mean-value multiplier system is generally indefinite. A single-node gauge must use symmetric elimination to preserve an SPD formulation. Code: C09.

CG 与重启 GMRES

T15-T16 / 根据算子性质选择克雷洛夫方法

[ ] T15 介绍 CG 的原理与适用条件

使用方法：Conjugate directions; Krylov spaces; energy minimization.

如何完成：On SPD B or a Dirichlet FEM stiffness K, use conjugate search directions to minimize the quadratic energy over growing Krylov spaces. Explain the sparse matrix-vector product and vector operations in each iteration, the modest storage cost, and why conditioning can slow unpreconditioned CG as the mesh is refined.

完成标志：CG principles, admissibility conditions and an energy-error bound; separate CG from PCG.

$$\|e_k\|_B \le 2[(\sqrt{\kappa_2}(B)-1)/(\sqrt{\kappa_2}(B)+1)]^k\|e_0\|_B$$

The inverse-preconditioner action in standard PCG must be fixed, linear and SPD. A general ILU or one-way GS action cannot automatically be paired with CG.

[ ] T16 介绍 GMRES 及其非对称适用场景

使用方法：Arnoldi orthogonalization; residual minimization; restarting.

如何完成：Build an Arnoldi basis and minimize the residual over its Krylov space. Apply restarted GMRES to original A and also to shared B as a control. Explain how restart length m affects basis storage, orthogonalization cost and possible stagnation.

完成标志：A GMRES(m) description and parameter table with inner steps, restarts and true residuals.

$$\text{Minimize } \|r_0-Gv\|_2 \text{ over } v\in K_k(G,r_0);\ G \text{ is A, B or K.}$$

预条件与停止准则

Fixed nonsymmetric preconditioners may be used with GMRES. Variable or nonlinear inner actions require a flexible method such as FGMRES, which is optional here. A library preconditioned-residual history does not replace the original-equation residual in T07.

Start with restart=30; test 20, 50 and 100 on a representative grid. General applicability does not imply lower cost. References [1]-[2] appear with the final selection guide.

GMG 与 AMG

T17-T18 / 校正不同尺度的误差

共同思想：平滑与粗网格校正
Local relaxation reduces oscillatory error; a coarse space corrects components that relaxation removes slowly. A V-cycle follows: pre-smooth → restrict the residual → solve recursively on the coarse level → prolong the correction → post-smooth.

[ ] T17 介绍几何多重网格 GMG

使用方法：Nested meshes; smoothing; coarse-grid correction.

如何完成：Use known geometry to build nested grids. Use bilinear prolongation on the regular square and nested-space prolongation for P1 FEM. Set homogeneous Dirichlet traces for error corrections. Specify restriction, coarse operators, smoothers and the coarsest solver.

完成标志：A hierarchy diagram and V-cycle specification, including the cost of building and maintaining geometric levels.

[ ] T18 介绍代数多重网格 AMG

使用方法：Algebraic aggregation; smoothed interpolation; Galerkin operators.

如何完成：Construct coarse spaces from the matrix graph and strong connections without a manually supplied geometric hierarchy. Start with smoothed aggregation and describe aggregation, candidate vectors, interpolation and smoothing. A constant candidate represents a near-null mode; it is not an exact null vector in the Dirichlet problem.

完成标志：A hierarchy specification recording coarsening, interpolation, smoothing and level sizes.

$$R=P^{T},\ B_c=P^{T}B_fP,\ u_f\leftarrow u_f+Pe_c$$

GMG obtains P from geometry or nested FEM spaces; AMG constructs P algebraically. Both can use Galerkin coarsening. GMG coarse matrices need not equal rediscretized matrices. This project first tests SPD configurations; nonsymmetric variants require separate choices.

Whether cycle counts remain approximately bounded under refinement, and whether either method is faster, are questions for measurement. References [3]-[4] support the component definitions.

多重网格独立求解与 PCG

T19 / 区分独立算法与组合方法

[ ] T19 为 PCG 规定固定对称 V 循环

使用方法：Symmetric multigrid; SPD preconditioning; PCG.

如何完成：On B, fix the hierarchy, smoothing counts and linear cycle action. Use $R=P^T$ and Galerkin coarse matrices. Pair forward-GS pre-smoothing with backward-GS post-smoothing, and use an SPD direct coarse solve. Explain why the resulting inverse-preconditioner action is symmetric positive definite.

完成标志：One validated V-cycle that can be tested independently and inside PCG.

角色	运行方式	统计内容
AMG / GMG 独立求解	Repeated residual corrections by V-cycles until accuracy is met.	V-cycles and work per level.
CG+AMG / CG+GMG	One fixed V-cycle per CG preconditioner call, starting from zero correction.	CG steps, V-cycle calls and total time.
固定非对称循环	Use a compatible outer method such as GMRES; do not assume PCG compatibility.	The cycle and outer iterations.
可变 / 非线性循环	Use an appropriate flexible method; optional in this project.	Inner changes and outer work.

$$C_{op}=\Sigma_\ell \mathrm{nnz}(B_\ell)/\mathrm{nnz}(B_0),\ C_{grid}=\Sigma_\ell n_\ell/n_0$$

These complexity measures describe hierarchy size. Actual memory also includes P, R, vectors and outer Krylov bases. One V-cycle is not the same amount of work as one CG step; iteration count alone cannot establish efficiency.

局部加密或非结构网格

AMG does not require a geometric hierarchy. GMG can participate when adaptive meshes retain reliable nested levels and transfers. Without these, mark that GMG implementation as not configured, rather than claiming that GMG cannot handle unstructured meshes.

四组实验分别回答什么问题

T20，第 A 部分 / S1-S4 的设计理由

Use controlled changes to connect performance to conditions. Within one matrix instance, all methods solve the same discrete problem. Across instances, change the size, operator representation, mesh hierarchy or number of right-hand sides deliberately.

场景	受控比较	要回答的问题
S1：规模	Four mixed-boundary combinations; N=16,32,64,128, then larger grids if feasible.	How do work, memory and time grow?
S2：系统表示	GMRES(A) versus GMRES(B), with the same physical discrete solution.	How much comes from row scaling rather than switching solvers?
S3：网格层级	L-shaped P1 FEM on uniform nested and local/adaptive meshes.	When do geometry and transfer construction favor GMG or AMG?
S4：重复利用	One fixed matrix; 1, 10 and 50 right-hand sides, reusing factors or levels.	When is setup cost amortized?

先保证同精度，再比较成本

With a manufactured solution, define $E_{disc}=\|U_{dir}-u^*\|$ and $E_{alg}=\|U_{method}-U_{dir}\|$. Require the common original-equation scaled residual and, as a suggested budget, $E_{alg}\le 0.01E_{disc}$. Check this in the relevant norm; for FEM, include continuous L^2 and energy norms.

$$T_{batch,solver}(R)=T_{setup}+\Sigma_{s=1\dots R}T_{solve,s}$$

Report setup and solve separately, together with peak memory and failures. For multiple right-hand sides, also report $T_{batch,solver}/R$. Assembly and row transformation are separate entries in end-to-end cost.

Fully converged methods should produce essentially the same discrete solution. The comparison concerns the resources and robustness needed to reach it. Rankings, crossover points and mesh-independent convergence are hypotheses to test, not acceptance targets.

按条件选择求解方法

T20，第 B 部分 / 先给实验证据，再提出建议

[ ] T20 用实验结果建立方法选择表

使用方法：Controlled experiments; matched accuracy; setup amortization.

如何完成：Combine matrix properties, mesh hierarchy, size, right-hand-side count and memory budget. Treat the guide below as a set of hypotheses. Use S1-S4 to support or revise each claim and state the tested range, failure cases and implementation choices.

完成标志：For each method: suitable conditions, measured evidence and practical limitations.

方法	待检验的适用场景	需要检查的限制
雅可比	Small systems, parallel smoothing or a teaching baseline.	Standalone convergence may be slow.
GS / SOR	Small/medium systems, favorable ordering and MG smoothing.	Ordering and relaxation-parameter dependence.
CG	Sparse SPD systems with limited basis storage.	Conditioning; original A is not directly admissible.
GMRES	Nonsymmetric systems and fixed general preconditioning.	Restarting, orthogonalization and basis memory.
AMG	SPD diffusion with no ready geometric hierarchy.	Setup cost and hierarchy complexity.
GMG	Regular grids or reliable nested geometric levels.	Transfers, boundary handling and local refinement.
稀疏直接法	Reference solves and repeated right-hand sides at manageable sizes.	Factor fill, peak memory and reuse cost.

参考资料： [1] SciPy CG; [2] SciPy GMRES; [3] PyAMG aggregation; [4] PETSc PCMG. The W and graph-energy proof are derived specifically for the stated discrete rows.